Report on the Value of House Sales in Upper Arlington

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Summary

- Recent house sales in Upper Arlington received $389 per ft^2 on average. Since 2014, house values in Upper Arlington increased 5.3% per year, which is at the low end of increases across communities in Franklin County (Grandview=+8%; Westerville = +7.2%).
- Evidence suggests that the two levies passed by Upper Arlington residents in 2017 and 2022 reduced growth in house values for a 1- to 2- year period relative to other districts in Franklin County. The effect of the 2017 tax increase was a 2.3% reduction in house value in 2019 (-$7.30/ft^2) and the effect of the 2022 tax increase was a 3.5% reduction in house value in 2024 (-$11.56/ft^2).
- The average price growth in Upper Arlington has been +$18.44 per ft per year over the last 2 years, or +36,145 per house per year for the average size sale (1960 ft^2).
- Living within a half mile of a middle school increases property value by $22,712 while living near an elementary school increases property value by $19,240, both compared to similar houses outside of a half mile radius.
- Living within 0.75 miles of the high school (but east of Northwest Boulevard and south of Fishinger road) reduces house value relative to other houses in Upper Arlington, and this loss grew after construction commenced in August 2021. This size of this effect is -4% to -7.6%, which is -$15.56/ft^2 to -$29.56/ft^2, or -$31,120 to -$59,128 for a 2000 ft^2 house within the 0.75-mile ring of the school. The effect is potentially larger for houses in a 0.5-mile ring, although fewer observations lowers the statistical significance of some of the estimates.

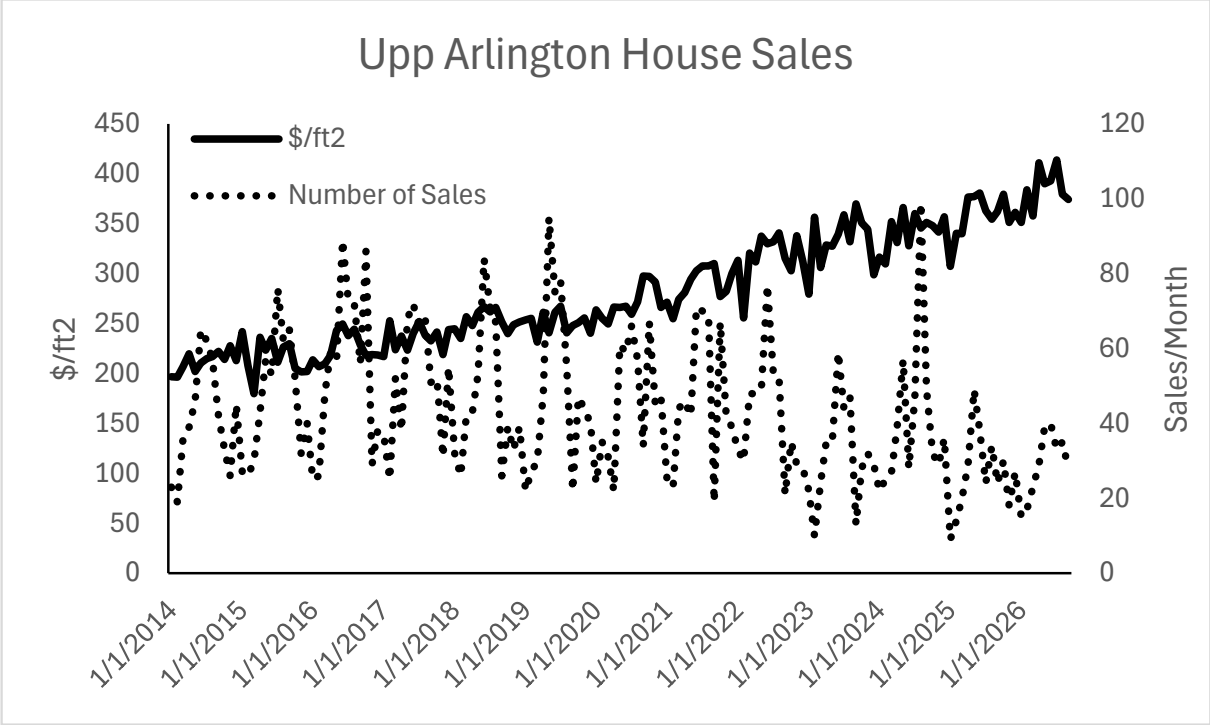
General Trends in House Values in Franklin County

This study reports on factors affecting the value of houses in Upper Arlington, Ohio. The results provide insights into what has happened to home values in the city since 2014 using analysis of over 190,000 home sales in Franklin County during that period, with more in-depth analysis of around 6000 home sales in Upper Arlington. The average sales prices of houses in Upper Arlington rose 5.3% per year during this 12-year period, with an average of 43 sales per month (Figure 1). The largest number of sales occurred in the month of June (around 62 sales per month) and the lowest number occurred in January or February. The average area of each home sold has stayed about the same, at around 1960 ft^2 per sale, meaning the average value per square foot has risen by nearly $190 per ft^2, from around $200/ft^2 in 2014 to $389/ft^2 in August and September of 2026.

On a $/ft^2 basis, Upper Arlington, Grandview and Bexley had the highest property sale values in the county (Figure 2). Over the 12-year period from 2014 to 2026, growth in house prices in Upper Arlington lagged the growth in house prices in other areas of Franklin County.

Grandview, Westerville, and unincorporated parts of Franklin County grew more quickly (e.g., prices in Grandview grew 8% per year). Price growth has slowed, and even reversed in some parts of the county, but in the last two years, trends have been stronger on the positive side in Upper Arlington than other parts of the county.

Figure 1: Number of house sales per month and average price of house sale, January 2014 to September, 2026. Data from Franklin County Auditor (N=6518)



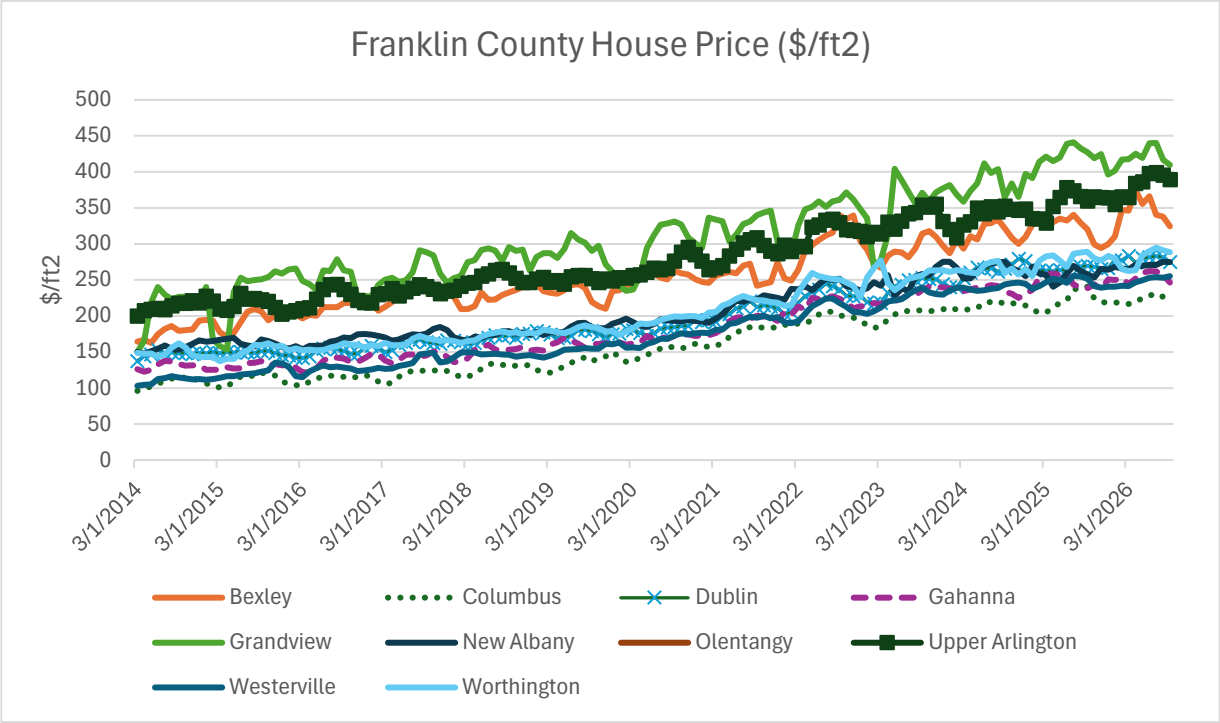
What is apparent from Figure 1 is the overall slowdown in house sales in Upper Arlington since 2021. This trend mirrors a reduction in sales across Franklin County and nationwide over the same period.

For homeowners in Upper Arlington, substantial changes in the community that could have influenced house values happened over the period from 2014 to 2026. These changes include two school levies, one in 2017 and another in 2022, and considerable construction around the schools and elsewhere. Among the districts shown in figure 2, Upper Arlington, Grandview and Bexley have the highest school tax rates, and for a brief period after the 2017 rate increase, Upper Arlington had the highest school tax rates. Rates increased in Grandview and Bexley in subsequent years, once again surpassing rates in Upper Arlington. Unlike the other districts in the region, New Albany experienced a reduction in school tax rates over this period.

School taxes implemented on property can have positive and negative effects on house value. On the negative side, taxes can reduce house value by increasing the costs of living. Effectively, the tax rate acts as an additional interest payment on ownership of a home, and interest rates and house value are inversely related. However, if the property tax enhances local services enough to increase demand for homes in the community, property values could rise. Such an effect could

be prominent in Upper Arlington, where the number of lots available for homes is limited. In this case, housing supply is inelastic and any perceived improvements in school quality that translate into increased housing demand will have strong positive effects on prices. The overall effect of school tax increase on house value will depend on the strength of these two factors.

Figure 2: Average price for house sales in selected Franklin County school districts, using a preceding 3 month moving average price. Top and bottom 1% of house sales by price dropped. Data from the Franklin County Auditor (N=130,842).



Below, figure 3 presents estimates from a regression analysis of 192,129 property sales in Franklin County between 2014 and 2026. The regression controls for characteristics of the house, location of the house, neighborhood amenities, and time of sale (The regression is presented in the Appendix). Using estimates of this model, I calculated the cumulative change in the value of the average house in Franklin County, Upper Arlington, and New Albany. Upper Arlington had two school related levies over the period analyzed, and in New Albany, school taxes decreased. Other changes happened in both communities, including the announcement of the new Intel facility near New Albany.

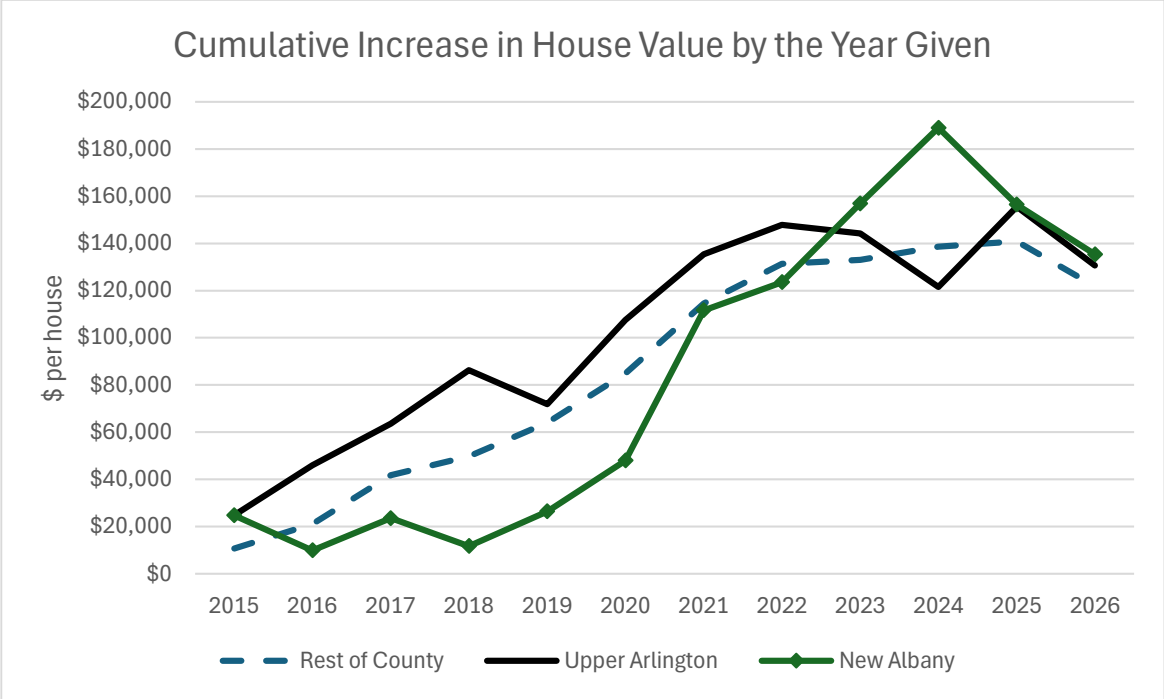
In general, Franklin County property values increased over the 12-year period, although price increases stabilized in the last few years and fell in the last year in the county. In Upper Arlington, the effect of the 2017 and 2022 school levies decreased house values both times relative to elsewhere in the county. Based on the values for Upper Arlington in Figure 3 and average house values in Upper Arlington over the 12-year period, I estimate that the effect of the 2017 tax increase was a 2.3% reduction in house value, or -$7.30/ft^2 in house value for the average house sale (1960 ft^2). For the 2022 tax increase, the reduction was 3.5%, or -$11.56/ft^2.

Importantly, prices did rebound in 2020 and 2025, but the two losses help explain the slower gains in property values in Upper Arlington in the past 12 years. Given the approach to estimation, the controls included, and the data available, it is unlikely that there are other explanations than the tax levies for the resulting house value declines in Upper Arlington in 2019 and 2023.

In New Albany, house values remained low from 2014 to 2019 but then began to rise. The effect of the Intel plant announcement in 2022 appears to have caused house value to increase considerably, although it already was on the upswing. House values have declined relative to other parts of the county there in the last two years, which is consistent with county-wide trends, although the effects in New Albany are especially pronounced.

One factor that has affected all homeowners in the past decade is national tax reform that altered the way home mortgage interest and local taxes could be treated on federal income taxes. This change in tax treatment had its most important effect on the largest and most valuable homes. However, this effect would be expected to influence all sales in Franklin County, not just the sales in Upper Arlington. It is unlikely that the lower prices experienced in Upper Arlington in 2018 and 2023 relate to the change in federal tax law. The most likely explanation for the decreases in house value in Figure 3 in Upper Arlington relate to things that affected Upper Arlington specifically.

Figure 3: Cumulative increase in house value by the year given. Numbers calculated for the average house. Estimates are based on regression equation (1) shown in the appendix.



Effects of Amenities in Upper Arlington

It is possible with this data to assess how amenities within Upper Arlington influence house values using more detailed analysis. In addition to tax reforms and changes in local tax rates, other changes have occurred in Upper Arlington in recent years, including large-scale redevelopment projects, for example along Lane Avenue, at various schools, and in the Kingsdale shopping center. These projects individually could influence nearby house prices.

The dataset used for this analysis includes most sales during that period except for the highest and lowest 1% of house sales in any given year by price and sales that had no land associated with them (e.g., condominiums and apartments). The regression analysis statistically determines how different amenities influence a home's value, including characteristics like square feet, bathrooms, lot size, distance from parks, distance from commercial sites, etc. The full regression is provided in the Appendix. The following tables group the amenities for Upper Arlington and provide estimates of the effect of each amenity on house values.

The data were downloaded from the Franklin County Auditor's website. This limits the number of house characteristics available compared to other analyses that may use data purchased from commercial providers. Commercial providers often obtain information and data from real estate agents through the MLS (Multiple Listing Service) system, aggregate sales over many years and sell the data.

In addition to data obtained from the Franklin County Auditor, a number of distance variables using straight line distances from the center of each property to the amenity in question. X and Y coordinates (latitude and longitude) of each parcel are provided for Franklin County in the Ohio South State Plane Coordinate System. The amenities included parks, schools, Scioto Country Club, the OSU Golf Course, the high school, and an identifier for properties on major roads, such as Henderson, Northwest Boulevard, West Lane Avenue, etc.

The first set of results in Table 2 focus on amenities of the house itself. Each additional ft^2 adds value to a home. In the regression model, house area is entered non-linearly, so the \$188 per ft^2 is the value at average area of the houses sold over the 12-year period, 1960 ft^2. The average size of houses sold in Upper Arlington remained about the same over the 12-year period, around 2000 ft^2. The results in the regression, however, do indicate that there is a premium placed on houses with greater than average finished area, with the value per ft^2 for a house with 3000 ft^2 being worth 21% more than the value of a house with 2000 ft^2. This effect helps explain why there are a significant number of rebuilds occurring in Upper Arlington, with a dual purpose to increase size and quality of the indoor space.

Bathrooms, bedrooms, and fireplaces, expectedly, increase home value, while the number of stories has a negative effect on home value. Age also has a negative effect, although the effect is not all that large. The average age of houses that sold in Upper Arlington is 70 years, and for the average house, an additional year of age reduces the value of the house by \$343.

The housing model includes some information on the type of build, for example, whether the structure is brick, stucco, siding, stone, or a mixture. The results in Table 2 indicate whether the

type of structure has a positive or negative impact on house value relative to the category of "mixture". Stone has a strong positive effect on house value, while siding has a negative effect. Both are significantly different from 0.

The value per acre of land is $230,784 per acre at the sample average of 0.30 acres per sale, meaning that on average land contributes $68,545 to each sale. The estimates on land suggest that the value of additional acres declines as lot size increases.

The timing of the house sale matters. House sales have their lowest value in the winter, but they rise and peak in June through August. If the sale occurred in March to May, it was on average $27,300 greater than a house that sold in December to February. If the house sold in June through August, it was $39,189 greater.

Table 2: Effect of house characteristics on house value. House values measured in real $, adjusted to 2026 to account for inflation. Significance level indicates statistical significance with *** implying a high likelihood the effect is > or < 0, and "-- " suggesting the result is not statistically different from 0. (N=6749)

Characteristic	**Effect $/unit**	**Significance**
House Area ($/ft2 measured at 2062 ft^2 avg)	$188	***
# Bathrooms ($/full bathroom)	$63,469	***
# Half Bathrooms ($/half bathroom)	$48,496	***
# Bedrooms ($/bedroom)	$16,355	***
# Fireplaces ($/fireplace)	$7,600	***
# Stories ($/story)	-$21,672	***
Age ($/yr measured at 67-year old average)	- $343	***
Siding* (wood, aluminum, or vinyl siding)	-$23,508	***
Brick* (brick)	-$6,592	--
Stucco* (stucco)	-$5,392	--
Stone* (stone)	$68,926	***
Acres ($/acre measured at 0.30 acre average)	$230,784	***
Average price growth ($/year)	$20,414	***
Sale occurred in Mar-May	$27,300	***
Sale occurred in Jun-Aug	$39,189	***
Sale occurred in Sep-Nov	$24,897	***

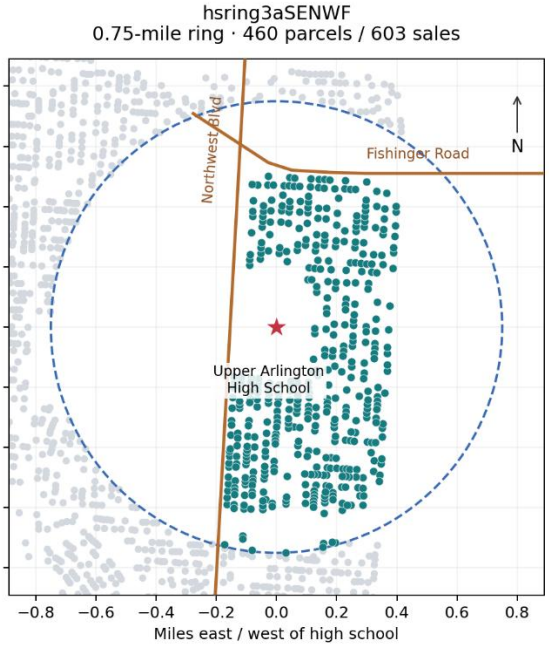
* additional value compared to base house type, which is partial frame/masonry (mixed)

Neighborhood characteristics affect house values in Upper Arlington as well. The Franklin County Auditor uses appraisal neighborhoods when considering value and assigns each house to a specific neighborhood. Variables accounting for these appraisal neighborhoods are included in our analysis but are not shown. Additional neighborhood characteristics related to the location of the property, and amenities within a given radius, are included in the analysis and shown in Table 3.

One of the most important developments in Upper Arlington in recent years has been the redevelopment of the high school lot. This, along with the redevelopment of Lane Avenue have likely influenced house values in the area north of Lane, east of Northwest Boulevard and South of Fishinger Road. Over the last 18 months, the area has been the site of additional construction related to a natural gas pipeline project that has seen numerous street closures, demolished homes, and new above- and below-ground pipeline infrastructure. The shifts in traffic patterns and parking associated with new traffic on Lane Avenue, the redeveloped high school, and the pipeline construction could affect house values.

The model presented in the Appendix assesses how the high school influences nearby housing values, where nearby houses are defined as those within a 0.75-mile distance, but east of Northwest Boulevard and south of Fishinger Road (Figure 4). In addition, the model tests whether the effect of the high school increased or decreased house value after construction began on the high school in 2021. The results of this test are summarized in Table 3. Before August, 2021, houses near the high school were valued $22,332 less than similar houses further than 0.75 miles away. After 2021, the negative impact of proximity to the high school grew by $28,768.

Figure 4: Treated parcels and sales (in aqua blue) contained in 0.75 mile high school ring east of Northwest Boulevard and South of Fishinger Road. Grey parcels are control parcels.



While the high school is a disamenity for neighbors, the elementary and middle schools in Upper Arlington are positive amenities. Families appear willing to pay a premium to live closer to these types of schools, with a stronger preference for proximity to a middle school.

Not surprisingly, proximity to the Scioto Country Club and generally being located south of Lane Avenue both have a positive effect on house value. Houses on main roads have slightly lower value, all else equal. Proximity to a park did not have a strong effect on house value.

Table 3: Effect of neighborhood characteristics on house value, with *** implying a high likelihood the effect is > or < 0, and "-- " suggesting the result is not statistically different from 0.

Characteristics	Effect ($/unit)	Significance
Within 0.75 miles of HS before 08/21 (E of NW, S. of Fishinger)	-$22,332	***
Within 0.75 miles of HS before 08/21 (E of NW, S. of Fishinger)	-$28,768	*
Within 0.5 miles of middle school	$22,712	***
Within 0.5 miles of elementary school	$19,240	***
Within 0.5 miles of Scioto CC	$133,475	***
On large road (e.g., Lane, NW)	-$33,126	***
On medium road (e.g., Mountview, Northam)	-$12,044	
South of Lane Avenue	$21,167	**

*Large roads= Northwest, Tremont, Fishinger, Lane Ave, Zollinger, Henderson, Redding, Reed.
*Medium roads=Mountview, Northam, Cambridge, Waltham, North Star

Tests of the High School Effect

The results in the regression model imply that the high school is a disamenity for nearby residents – an outcome that contrasts with the apparent benefits of living near middle and elementary schools in Upper Arlington. What is more surprising about the high school effect is that it appears to have become more negative during and after construction on the new high school began in August of 2021.

To analyze this effect more carefully, I conduct additional analysis to test directly whether house values inside a ring around the Upper Arlington High School fell after construction began in 2021. For this analysis, houses inside the ring above are labelled "Treatment" and houses outside the ring are labelled "Control." I also test a smaller 0.5 mile ring.

Each of the models tests the following equation

$$\text{Difference} = (\text{House value before 2021}_{\text{Treatment}} - \text{House value before 2021}_{\text{Control}}) - (\text{House value after 2021}_{\text{Treatment}} - \text{House value after 2021}_{\text{Control}})$$

A negative value for this difference indicates that the house value inside the ring has fallen relative to house values outside the ring. All house values are like to be increasing over time, so

this set of tests just analyzes the relative shift in house values before and after 2021 inside the ring (treated) and outside the ring (controls).

To conduct this analysis, I downloaded the original data for 2014 to 2026 from the Franklin County Auditor website and created a dataset consisting of 6750 house sales in Upper Arlington over this period. I then used ChatGPT to code and run these analyses in Python as well Stata on my own computer. This analysis is described in more detail in the appendix under "regression 3." The data and code are available at https://github.com/bsohngen/UpperArlington2026.

Two sets of controls – observations outside of the ring around the high school – are used in this analysis: (1) all observed sales in Upper Arlington and (2) sales within an additional 0.75 miles outside the ring. I also use matching in the analysis to ensure house characteristics in the control data are similar to the characteristics in the treatment data.

The results are shown in Tables 4 and 5 for the two rings (0.75 mile and 0.5 mile). The "conventional" rows are derived from a similar analysis as presented in Appendix Table 2. A different set of regressors was used here focusing explicitly on the housing characteristics but not the neighborhood amenities. In addition, monthly time effects are used here rather than annual time effects as used in Appendix Table 2. The column for "percent effect" is derived from a log model while the column for the "Dollar effect" is derived from a linear model. No matching is used in the conventional analysis.

The Doubly Robust approach uses the method described in Sant'Anna and Zhao (2020)[1]. To interpret the parameters in Tables 4 and 5 as causal, i.e., the high school and subsequent construction caused the shift in relative prices, the "parallel trends" assumption should hold. This assumption means the trend in house values inside the ring and outside the ring do not differ before 2021. Tests of this assumption generally do not reject this assumption (i.e., the assumption generally holds), however, one test under the matching approach described below did reject the parallel trends assumption.

The matching models use a propensity score model to predict a set of parcels outside of the ring that essentially look like parcels inside the ring. The same regression as used in the conventional analysis is conducted, however, fewer observations are used (1894 sales for the 0.75 mile ring and 1449 sales for the 0.5 mile ring) because the approach only uses sales on parcels outside the ring that are good matches for sales on parcels inside the ring.

All the effects in Table 4 are negative, however, several of them are not statistically significant. For the conventional model, the parameters in the log regression (percent effect) are not significant. For the doubly robust method, the linear dollar effect is not significant for the "All" sample. The percent effect in the doubly robust approach is fairly large, indicating a potential relative price decline inside the 0.75 mile ring after 2021 of 4.0% to 7.6%.

Significance generally declines for the analyses of the 0.5-mile ring. The effects in the conventional models get smaller and their significance is similar. In contrast, the effects in the

[1] Sant'Anna, P.H. and Zhao, J., 2020. Doubly robust difference-in-differences estimators. *Journal of econometrics*, *219*(1), pp.101-122.

doubly robust models become larger and they have the same pattern of significant results with all models suggesting a significant relative decline in property values inside the ring except for the linear (dollar effect) doubly robust "All" sample. The potential percentage change in house value for this ring based on the doubly robust approach is -4.8% to -10%. The matching models are all insignificant and one parameter is positive, albeit small.

Table 4: Change in relative real (adjusted for inflation) house value for treated houses within a three-quarter mile ring (east of Northwest and south of Fishinger) of the Upper Arlington High School after August 2021 compared to houses outside of the ring. "All" controls = controls drawn from all sales outside of the ring; "Nearby" controls drawn from the ring 0.75 to 1.5 miles from high school.

Method and controls	Dollar effect	p value	Percent effect	p value
Conventional All	- $37,691	0.002	-1.47%	0.398
Conventional Nearby	- $36,731	0.008	-1.52%	0.429
Doubly Robust All	- $10,563	0.328	-3.97%	0.030
Doubly Robust Nearby	- $44,644	0.006	-7.56%	<0.001
Matched All	- $29,046	0.032	-3.22%	0.104
Matched Nearby	- $27,198	0.076	-2.69%	0.242

Table 5: Change in relative house value for treated houses within a half mile ring (east of Northwest and south of Fishinger) of the Upper Arlington High School after August 2021 compared to houses outside of the ring. "All" controls = controls drawn from all sales outside of the ring; "Nearby" controls drawn from the ring 0.75 to 1.5 miles from high school.

Method and controls	Dollar effect	p value	Percent effect	p value
Conventional all	$-27,124	0.043	-0.35%	0.852
Conventional nearby	$-28,989	0.065	-0.41%	0.846
Doubly robust all	$-14,379	0.250	-4.79%	0.018
Doubly robust nearby	$-67,901	0.004	-9.95%	<0.001
Matched all	$-2,824	0.846	0.08%	0.970
Matched nearby	$-20,426	0.327	-1.57%	0.527

Conclusion

This study examined house values across Franklin County from 2014 to September 2026. Upper Arlington is among the communities with the highest housing values in Franklin County (recent sales around $\$389/ft^2$), but its price growth in the past 12 years has lagged most other parts of the county. Price growth was 5.3% per year from 2014 to the present. Only growth in New Albany

was lower although their struggles have come mainly in the past two years. Note that this analysis includes only parcels in Franklin County for communities that straddle two counties like New Albany, Dublin, and Westerville.

When looking at the trends over time, the data suggest that the two recent tax levies passed by Upper Arlington residents in 2017 and 2022 had negative effects on price growth, slowing it significantly for 1-2 years after the passage of the levies. Given this effect, the levies potentially explain the relative lag in value Upper Arlington residents have experienced in house value growth over the 12-year period.

Detailed analysis of house sales in Upper Arlington provides some additional information on what drives value in the community. Not surprising to any Upper Arlington residents, location matters considerably. South of Lane and proximity to the Scioto Country Club both substantially increase value. So do proximity to a middle school or elementary school. However proximity to the high school has a negative effect on house value.

This effect surprised me so I investigated it further with more detailed analysis to better understand what has happened. The initial regression analysis suggested that not only are houses near the high school devalued relative to other parts of the city, but the devaluation strengthened after construction on the new high school commenced. To better understand this, I conducted a series of additional statistical analyses that focused on whether house values had potentially declined in a ring around the high school after 2021 (construction began in August, 2021).

The implication of these analyses suggest that house values fell inside a 0.5 or a 0.75 mile ring around the high school after 2021 relative to houses outside the high school ring. This does not mean house values fell absolutely, it means they did not gain value as quickly as houses outside the ring. One result suggested the effect could be as large as a negative 10% for houses in a 0.5-mile ring. Not all tests were statistically significant, however the strength of the tests combined indicates to me that houses values around the high school are devalued relative to similar houses further from the high school and this effect got bigger after construction.

Appendix: Regression Results

Regression 1: Franklin County Model

The Franklin County regression analysis is based on 192,129 total sales from 2014 to 2026. I conducted a linear regression analysis in SAS using the Generalized Linear Model (GLM) command. I included fixed effects for neighborhoods, as defined by the county auditor, and time periods. The neighborhood fixed effects are suppressed and not shown. The annual fixed effects are listed as d15 to d26. The data begin in 2014, so a fixed effect for 2014 is left out of the model.

The parameters on d15 to d16 represent the cumulative change from 2014 to the year of the parameter estimate in house value for the average house outside of Upper Arlington or New Albany in Franklin County. To determine the cumulative change for a home in Upper Arlington, add "d15 + uad15" for 2015, or "d20 + uad20" for 2020. For New Albany, make the same calculation (e.g., the New Albany effect for 2017 is "d17 + newad17"). These are the values presented in Figure 3 above.

To determine statistical significance, the typical standard is that the parameter is statistically significant if Pr>|t| is smaller than 0.05.

Table A1: Parameter estimates for Franklin County regression. Generalized Linear Model with neighborhood fixed effects.

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Area (ft^2)	83.4515	0.9404	88.74	<.0001
Baths (#)	35342.8	641.46138	55.1	<.0001
Half baths (#)	20770	675.05029	30.77	<.0001
bedrooms	4360.08	559.31863	7.8	<.0001
Fireplace (yes=1; no=0)	1400.91	542.09756	2.58	0.0098
Stories (#)	-1410.8	81.64898	-17.28	<.0001
Year built	248.896	35.50826	7.01	<.0001
Wood exterior	226.064	719.25222	0.31	0.7533
Brick	9367.44	1481.24251	6.32	<.0001
Stucco	-245.86	1636.51388	-0.15	0.8806
Distance from county center	-0.1448	0.13253	-1.09	0.2746
d15 (2015)	10680.3	1405.36346	7.6	<.0001
d16 (2016)	21137.8	1341.26691	15.76	<.0001
d17 (2017)	41824.7	1347.96076	31.03	<.0001
d18 (2018)	49733.1	1383.58482	35.95	<.0001
d19 (2019)	64019.9	1385.09822	46.22	<.0001

d20 (2020)	84936.9	1399.20067	60.7	<.0001
d21 (2021)	114499	1375.41745	83.25	<.0001
d22 (2022)	131279	1406.31065	93.35	<.0001
d23 (2023)	133032	1506.36589	88.31	<.0001
d24 (2024)	138587	1724.32426	80.37	<.0001
d25 (2025)	140793	1891.26762	74.44	<.0001
d26 (2026)	121766	2367.85611	51.42	<.0001
Repeat sale (1,0)	4055.5	535.69064	7.57	<.0001
Longitude	-0.024	0.16398	-0.15	0.8835
Latitude	-0.3136	0.24754	-1.27	0.2052
Sale in 2nd quarter	4605.5	732.86462	6.28	<.0001
Sale in 3rd quarter	9500.22	692.20138	13.72	<.0001
Sale in 4th quarter	10345.2	744.16059	13.9	<.0001
uad15	14216.5	5586.47926	2.54	0.0109
uad16	24783.3	5389.31728	4.6	<.0001
uad17	21862.6	5401.23965	4.05	<.0001
uad18	36485.6	5534.49673	6.59	<.0001
uad19	7882.77	5477.39264	1.44	0.1501
uad20	22668.8	5571.05236	4.07	<.0001
uad21	20833	5554.95775	3.75	0.0002
uad22	16505	5659.53772	2.92	0.0035
uad23	11166.1	6316.44629	1.77	0.0771
uad24	-17056	6666.01111	-2.56	0.0105
uad25	14755.4	7616.88721	1.94	0.0527
uad26	8888.78	8467.62589	1.05	0.2938
newad15	14054.6	6977.75669	2.01	0.044
newad16	-11235	6531.97359	-1.72	0.0854
newad17	-18263	6680.40629	-2.73	0.0063
newad18	-37957	7055.974	-5.38	<.0001
newad19	-37593	7035.7988	-5.34	<.0001
newad20	-36953	7257.14568	-5.09	<.0001
newad21	-2857.5	7082.20942	-0.4	0.6866
newad22	-7722.6	8223.53366	-0.94	0.3477
newad23	23895.9	9639.65349	2.48	0.0132
newad24	50363	10829.2664	4.65	<.0001
newad25	15614.3	11955.3202	1.31	0.1915
newad26	13611.4	13563.2739	1	0.3156

Regression 2: Upper Arlington Model

The Upper Arlington regression analysis is based on 6749 sales from 2014 to 2026. I conducted a linear regression analysis in SAS using the Generalized Linear Model (GLM) command. I included fixed effects for neighborhoods, as defined by the county auditor, and time periods. The neighborhood fixed effects are suppressed and not shown. The annual fixed effects are listed as d15 to d26. The data begin in 2014, so a fixed effect for 2014 is left out of the model.

The parameters on d15 to d16 represent the cumulative change from 2014 to the year of the parameter estimate in house value for the average house in Upper Arlington. These parameters show a similar pattern as the Upper Arlington parameters in Figure 3. House values have generally increased, but experienced two significant reductions after the two property tax increases (2017, 2022)

As in regression 1 above, the typical standard to determine statistical significance is that the parameter is statistically significant if Pr>|t| is smaller than 0.05.

Table A2: Parameter estimates for Franklin County regression. Generalized Linear Model with neighborhood fixed effects.

Parameter	Estimate	Standard Error	t Value	Pr > |t|
Area (ft^2)	107.5596	9.42958	11.41	<.0001
Area*Area	0.0191	0.0013	14.63	<.0001
Baths (#)	63468.9026	3405.54236	18.64	<.0001
Half baths (#)	48495.9063	3812.87092	12.72	<.0001
bedrooms	16355.3596	3408.02814	4.8	<.0001
Fireplace (yes=1; no=0)	7599.6926	2362.7685	3.22	0.0013
Stories (#)	-21671.524	6070.21932	-3.57	0.0004
Year built	-302040.7	17593.22687	-17.17	<.0001
(Year built)*(Year built)	77.3147	4.489	17.22	<.0001
Wood exterior	-23507.674	5513.19801	-4.26	<.0001
Brick	-6592.3652	6952.19626	-0.95	0.343
Stucco	-5391.5769	8380.22173	-0.64	0.52
Stone	68925.5118	19839.30692	3.47	0.0005
Acres	255508.475	28521.01639	8.96	<.0001
Acres*acres	-83245.846	16104.66707	-5.17	<.0001
0.75 mile high school ring (S of Fishinger/E of NW)	-22331.503	9059.72146	-2.46	0.0137

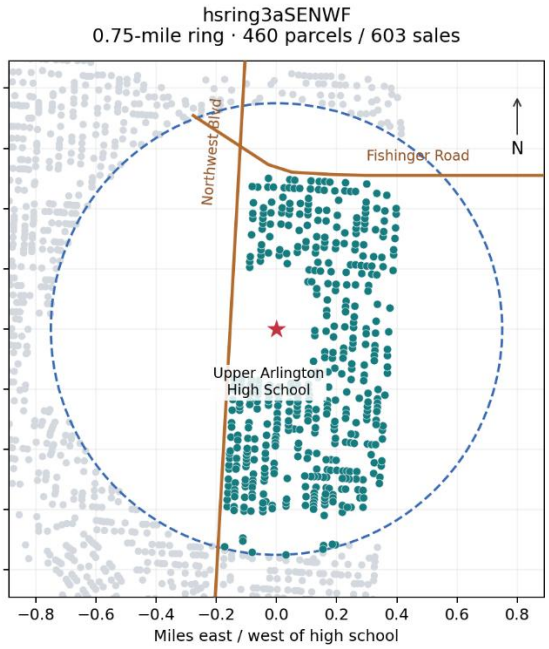
(0.75 mile high school ring)*(After 08/2021)	-28767.6	15759.52288	-1.83	0.068
Distance to park (miles)	0.0067	0.02357	0.28	0.7779
W/in 0.5 mile of Scioto CC	133475.335	10791.11788	12.37	<.0001
W/in 0.5 mile of middle school	22711.8925	5663.35167	4.01	<.0001
W/in 0.5 mile of elementary school	19239.8908	4416.24686	4.36	<.0001
Repeat sale in 12 yr period	22146.533	3717.09599	5.96	<.0001
House on large road	-33126.846	7203.14296	-4.6	<.0001
House on medium size road	-12044.136	12389.58909	-0.97	0.331
d15 (2015)	16501.0988	9128.61693	1.81	0.0707
d16 (2016)	37066.3278	8874.20909	4.18	<.0001
d17	58000.6688	9039.551	6.42	<.0001
d18	83149.3756	9267.5574	8.97	<.0001
d19	77301.2186	9125.47216	8.47	<.0001
d20	112120.519	9247.56898	12.12	<.0001
d21	140477.346	9172.8324	15.31	<.0001
d22	178962.066	9407.62507	19.02	<.0001
d23	182513.702	10075.23615	18.12	<.0001
d24	183385.023	10603.66096	17.29	<.0001
d25	221369.34	11193.48196	19.78	<.0001
d26	244964.578	11994.08791	20.42	<.0001
House south of Lane	21166.8097	8585.50584	2.47	0.0137
Sale in 2nd quarter	27300.3163	5720.08621	4.77	<.0001
Sale in 3rd quarter	39188.8606	5403.46537	7.25	<.0001
Sale in 4th quarter	24896.5513	5948.28829	4.19	<.0001

Regression 3: High School Perimeter Analysis

This part of the appendix provides an explanation of the detailed analysis about whether property values are depressed around the the high school relative to other areas of Upper Arlington and whether the difference in property values near the high school compared to further away expanded after August, 2021 when construction on the new high school began.

This analysis was conducted primarily for a 0.75 mile ring around the high school that excluded properties west of Northwest Boulevard and north of Fishinger Road. The parcels in aqua blue in Figure 4 below are the treated parcels, where the treatment is being near the high school. Control parcels are in grey. Note that additional controls in this analysis are outside the box shown.

Figure 4 (from main text): Treated parcels and sales (in aqua blue) contained in 0.75 mile high school ring east of Northwest Boulevard and South of Fishinger Road. Grey parcels are control parcels.



This guide explains the analysis included in "UA_Both_Rings_Complete_Package.Zip" found on the following Github site: https://github.com/bsohngen/UpperArlington2026

The analysis includes conventional difference in differences (DID), dedicated doubly robust DID, and matching followed by DID. For the analysis, I assessed results for two restricted school rings, the 0.75 mile ring above and a 0.5 mile ring over two control groups (all of the rest of Upper Arlington outside of the 0.75 mile ring, and a control group composed of houses outside the 0.75 mile ring but within 1.5 miles of the school). Below are discussion of the model equations, results, and diagnostic checks. Instructions for using the code can be found at the Github site.

Key findings: Doubly robust log-price estimates suggest relative declines after September 2021. Conventional and matched log-price estimates are not statistically significant. One of the specifications – the three-quarter-mile nearby matched specification – rejects the pre-trend test. Overall the results imply that house values inside either ring boundary fell after construction began, but the rejection of the pre-trend test limits the ability of this analysis to conclude that the construction caused this reduction in house value. That is, house values fell inside the ring after August 2021, but there is a possibility that the decline could have been caused by something other than the construction itself.

Table A3: Results included in the analysis

Analysis	Half mile ring	Three quarter mile ring
Conventional DID and event study	Included	Included
Doubly robust DID	Included	Included
Matching followed by DID	Included	Included

The half-mile flag is "hsring2aSENWF"; the three-quarter-mile ring is "hsring3aSENWF". Both restrict ring membership to parcels east of Northwest Boulevard and south of Fishinger Road. Their "after" interactions (sales after August 2021) are "hsring2aSENWFafter" and "hsring3aSENWFafter".

The question being estimated

Each method asks whether prices inside the restricted ring changed after September 2021 relative to comparison prices. A negative effect on "hsring2aSENWFafter" and "hsring3aSENWFafter" means less favorable price appreciation inside the ring. It does not necessarily mean inside prices fell in absolute terms. If inside prices increase by $80,000 and control prices increase by $110,000, the relative effect is −$30,000.

Data covariates and comparison groups

The source is the uploaded ua_all_cpi3d4.csv with 6,761 sales. Complete-case analysis retains 6,750 sales from 4,994 parcels and excludes 11 observations. A parcel sold more than once contributes more than one sale; parcel clustering or resampling accounts for dependence among its sales. No new correction to stories or other supplied values is imposed.

Let D equal 1 inside the selected restricted ring and 0 for controls. Let A equal 1 for September 2021 onward and 0 earlier. Y is "realp" or "lnrealp". X contains the eleven covariates. The ring-after field is checked to equal D multiplied by A.

realp is the CPI-adjusted sale price supplied in the dataset, using 2026 as the base year. lnrealp is its natural logarithm. Percentage effects in the result tables transform log coefficients as follows:

$$\text{Percent effect} = 100(e^{\hat{\tau}_{\log}} - 1)$$

All models use area_a, baths, hbaths, bedrms, firep, stories, yrblt, wdalvn, brick, stucco, and acres. These describe house size, full and half bathrooms, bedrooms, fireplaces, stories, construction year, exterior materials, and lot size. Stata scales area_a by 1,000 and centers yrblt at 1950. These changes do not alter the fitted model space. Neither neighborhood nor distance is an additional covariate.

Table A4: Treated sales numbers.

Treated sales	Before	After	Total
hsring3aSENWF (0.75 mile ring)	381	201	582
hsring2aSENWF (0.5 mile ring)	283	153	436

All controls are every complete-case sale whose selected restricted-ring flag is zero. This includes properties in portions of the original circle excluded by the road restrictions. For the half-mile ring, all controls also include properties between 0.5 and 0.75 mile.

Nearby controls are outside the original 0.75-mile circle and within 1.5 miles of the school. The same geographic band is used for both rings. Thus, the nearby half-mile analysis excludes the 0.5–0.75-mile band from its controls. All eligible treated sales are retained before matching.

Table A5: Total sales/parcels. Sales>Parcels implies repeat sales in data.

Conventional and DR sample	Sales	Parcels
All controls either ring	6750	4994
Nearby controls three quarter mile	3223	2346
Nearby controls half mile	3077	2236

Conventional difference in differences (DID)

The result label DID_month_FE denotes an unweighted regression with sale year-month fixed effects. It does not perform matching nor does it perform inverse probability weighting. This is similar to regression 2 in the Upper Arlington analysis of this paper, however, it does not include neighborhood characteristics. Without covariates, the basic DID contrast is:

$$\hat{\tau}_{DID} = (\overline{Y}_{1,1} - \overline{Y}_{1,0}) - (\overline{Y}_{0,1} - \overline{Y}_{0,0})$$

The first subscript identifies treated or control, and the second identifies before or after. The regression adjusts that contrast for housing characteristics and common monthly price movements:

$$Y_i = \alpha + \beta D_i + \tau D_i A_i + X_i'\gamma + \lambda_{m(i)} + \varepsilon_i$$

Beta (β) is the adjusted pre-period ring price gap. Tau (τ) is the change in that gap after September 2021. Lambda (λ) is a fixed effect for each sale year-month. A separate after indicator is absorbed by those time effects. The model imposes common covariate slopes across groups and periods.

For the three-quarter-mile ring with all controls, β in the dollar model is −$3,950 and τ is −$37,691. The fitted post-period gap is their sum, approximately −$41,641. The DID effect is the change in the gap, not the full post-period gap.

The main inference label parcel means standard errors are clustered by parcel. The spatial_1000ft sensitivity instead clusters sales by a 1,000-foot geographic grid. These choices change standard errors rather than point estimates. The DID_month_FE_exclude2021 sensitivity removes all 2021 sales to check dependence on the transition year.

Annual event study

The event study replaces the single ring-after term with ring interactions for annual periods. It retains covariates and sale year-month effects, uses 2020 as the reference, and splits 2021 into January–August and September–December. Each coefficient measures the relative ring gap in that period compared with its gap in 2020.

The pretrend test jointly tests the 2014–2019 and January–August 2021 interactions. A small p value signals evidence against stable pre-period relative gaps. A large p value does not prove parallel trends; the test can lack power and does not directly validate a different DR specification.

Doubly robust DID with inverse probability weights

DRDID_RC_locally_efficient model denotes a dedicated repeated-cross-section doubly robust DID estimator following Sant Anna and Zhao (2020)[1] equation 3.4. Repeated cross sections means that before and after sales need not involve the same properties. Locally efficient refers to an efficiency property under correctly specified models and the estimator's assumptions.

An unweighted pooled logistic regression predicts ring membership from eleven covariates. The model includes an intercept. Controls receive odds weights derived from the estimated probabilities:

$$p(X_i) = P(D_i = 1 \mid X_i), \qquad w_i = \frac{\widehat{p}(X_i)}{1 - \widehat{p}(X_i)}$$

Controls with characteristics more typical of ring properties receive greater weight. The weights are normalized within each period when calculating control averages. They do not determine whether a control is matched or selected.

For each outcome, the code fits four unweighted linear outcome regressions: controls before, controls after, treated before, and treated after. Define their conditional predictions as m(d,a)(X). Separate fits allow the relationships between characteristics and prices to vary across group and period. Propensity weights enter residual averages, not these regression fits.

$$R_{i,a} = Y_i - \widehat{m}_{0,a}(X_i)$$

$$G_a = \overline{R}_{D=1, A=a} - \frac{\sum_{D_i=0, A_i=a} w_i R_{i,a}}{\sum_{D_i=0, A_i=a} w_i}$$

G(a) is the treated residual mean minus the weighted control residual mean. The before/after contrast is G(1) minus G(0). The locally efficient version also adjusts to the pooled treated covariate distribution:

$$q_a(X) = \widehat{m}_{1,a}(X) - \widehat{m}_{0,a}(X)$$

$$C_a = \overline{q_a(X)}_{D=1} - \overline{q_a(X)}_{D=1, A=a}$$

$$\hat{\tau}_{DR} = (G_1 - G_0) + (C_1 - C_0)$$

This correction standardizes the model component to all treated sales pooled across periods under the repeated-cross-section assumptions. All averages are over sales, so repeatedly sold parcels contribute multiple observations.

The estimator is doubly robust: under the identification assumptions it can remain consistent when either the propensity model or the required control outcome-change model is correct. It still requires conditional parallel trends, overlap, no anticipation, and appropriate stable sampling assumptions. It does not include the conventional model's full year-month effects.

Matching followed by DID

Matching_then_DID is a robustness analysis for both restricted rings. It first estimates an unweighted pooled propensity model using the eleven covariates, then matches each treated sale to up to three nearest control sales within the same before/after period. Matching uses predicted logit-score distances and a caliper of 0.2 times their pooled sample standard deviation. Outcomes are not used to select matches.

Controls may be reused. A retained treated sale has weight one. A control's matching weight is the sum of 1/k for treated sales to which it is matched, where k is that treated sale's number of selected controls. These are matching weights, not inverse probability weights. Matching is performed at the sale level rather than once per parcel.

The selected observations are then used in the conventional DID equation with the eleven covariates and month fixed effects, weighted by their matching contributions. All 582 three-

quarter-mile and 436 half-mile treated sales receive matches. The all-control matched sample contains 1,894 sales from 1,577 parcels; the nearby matched sample has 1,544 sales from 1,238 parcels. Half-mile matching retains 1,449 sales from 1,230 parcels with all controls and 1,211 sales from 984 parcels with nearby controls.

Raw and matched standardized differences

$$\text{SMD} = \frac{\overline{X}_T - \overline{X}_C}{\sqrt{(s_T^2 + s_C^2)/2}}$$

raw_smd compares group means before matching. matched_smd compares selected treated and control means using matching weights. Both use the same raw period-specific pooled standard deviation, so the change reflects mean balance. Absolute values below 0.10 are a common heuristic, not a guarantee of valid causal identification. These results indicate that the matches are good, but not great.

Table A6: SMD = Standardized Mean Differences, a measure of difference between the treated and control groups. Before maximum absolute SMD is the maximum mean difference for the before period while after is the maximum mean difference in the after period.

Matched balance	Before maximum absolute SMD	After maximum absolute SMD
Three quarter mile all	0.117	0.108
Three quarter mile nearby	0.156	0.082
Half mile all	0.076	0.146
Half mile nearby	0.141	0.145

Matching improves overall balance but leaves some covariates above 0.10. Balance concerns remain even when propensity scores are close. The matched nearby log-price pretrend test also rejects stable pre-period gaps, as reported in the diagnostic table.

Main estimates for both restricted rings

These tables report the verified Python baseline results using parcel-clustered conventional inference, parcel-bootstrap DR inference, and conditional parcel-clustered matched inference. Dollar effects come from realp; percentage effects transform lnrealp coefficients. Their p values test their respective model coefficients against zero.

Table A7: Results for three quarter mile ring (0.75)

Method and controls	Dollar effect	p value	Percent effect	p value
Conventional all	$-37,691	0.002	-1.47%	0.398
Conventional nearby	$-36,731	0.008	-1.52%	0.429
Doubly robust all	$-10,563	0.328	-3.97%	0.030
Doubly robust nearby	$-44,644	0.006	-7.56%	<0.001
Matched all	$-29,046	0.032	-3.22%	0.104
Matched nearby	$-27,198	0.076	-2.69%	0.242

Table A8: Results for half mile ring (0.5)

Method and controls	Dollar effect	p value	Percent effect	p value
Conventional all	$-27,124	0.043	-0.35%	0.852
Conventional nearby	$-28,989	0.065	-0.41%	0.846
Doubly robust all	$-14,379	0.250	-4.79%	0.018
Doubly robust nearby	$-67,901	0.004	-9.95%	<0.001
Matched all	$-2,824	0.846	0.08%	0.970
Matched nearby	$-20,426	0.327	-1.57%	0.527

For both rings, DR log-price estimates are negative and statistically significant with either control group. Conventional log estimates are not significant. Matching for both rings also gives insignificant log estimates. Half-mile matched dollar estimates are also insignificant. Dollar and log models need not agree because they emphasize price differences differently.

The methods are not interchangeable. They differ in weights, outcome-model flexibility, selected sample, composition adjustment, and time controls. The DR method has no full month fixed effects. A smaller p value is not evidence that a model is necessarily better.

Uncertainty in pretrends and interpretation

Each DR analysis uses 499 parcel bootstrap draws. A draw selects the original number of parcel clusters with replacement and includes all their sales. A parcel selected twice contributes its sales twice. For two outcomes, every draw refits one propensity model and eight outcome regressions. All 499 fits succeeded for both rings and control groups.

All-control draws select 4,994 clusters. Nearby draws select 2,346 for the three-quarter-mile ring and 2,236 for the half-mile ring. The sales count varies with the drawn parcels, and the number of distinct parcels is smaller because of replacement. Bootstrap standard errors are the standard deviations of the estimates. Reported Python DR p values and intervals use a normal approximation.

Matched DID standard errors condition on the fitted matches. They do not fully account for estimating matches or cross-parcel dependence induced by matching. Its p values are approximate robustness diagnostics. The package does not use an ordinary nearest-neighbor matching bootstrap. Parcel inference across all methods also does not fully resolve correlation among neighboring parcels.

Table A9: Joint log price pretrend tests

Specification	All controls p value	Nearby controls p value
Conventional three quarter mile	0.495	0.626
Conventional half mile	0.833	0.831
Matched three quarter mile	0.384	0.033
Matched half mile	0.702	0.198

The three-quarter-mile matched nearby sample rejects at the 5% level, weakening its DID interpretation. Other tests do not reject, but non-rejection does not prove the counterfactual trends would be parallel. DR inference depends on the identification assumptions in addition to the estimated models.

$$E[Y_1(0) - Y_0(0) \mid D = 1, X] = E[Y_1(0) - Y_0(0) \mid D = 0, X]$$

Conditional parallel trends means comparable ring and control properties would have followed the same untreated price change. Y(0) denotes the outcome without the intervention. Anticipation, changing sale composition, local redevelopment, spillovers onto controls, and weak overlap can undermine causal interpretation.

The appropriate conclusion is evidence of relative declines under the DR specifications with sensitivity to the adjustment method. Prices could have increased inside the ring while increasing

less than controls. These estimates alone do not establish that the school caused a decrease in property values.

Method reference: Sant Anna and Zhao (2020), Doubly robust difference in differences estimators, Journal of Econometrics 219, 101–122.

Running the package and locating outputs

Stata entry point

Extract the ZIP and keep its folder structure. Open run_all.do, edit the cd line to the extracted UA_complete_package folder, and run the entire file. It calls stata/UA_DID_updated.do, stata/UA_ring2_DID.do, stata/UA_matched_DID.do, and stata/UA_matched_ring2_DID.do. No user-written Stata packages are required. Outputs are saved in stata_results, including All_main_results.csv.

The conventional modules define covariates immediately before each regression section; the DR program defines them internally. If conventional output lists only the ring and interaction, the intended covariates were omitted. Full output should list the eleven covariates. Run whole files so local macro definitions are available.

Python entry point

From the package folder, install dependencies with python -m pip install -r requirements.txt. Run python run_all.py. The master script runs all methods and writes python_results/All_main_results.csv and separate method folders. The default is 499 bootstrap replications and seed 20261007; --reps and --outdir change these options.

The combined Python master run reproduced the baseline estimates and standard errors. Stata code was not executed here; the user's earlier all-control DR run matched the Python point estimates. Bootstrap standard errors can differ slightly across languages because the random draws differ. Matching can differ if estimated scores have nearly tied distances.

Files to inspect

results/All_main_results.csv combines the main estimates. The hsring3a and hsring2a folders contain DID_results.csv, Regression_coefficients.csv, Event_study_results.csv, Pretrend_tests.csv, Sample_counts.csv, bootstrap draws, and event-study figures. The matching and matching_hsring2a folders contain results, raw and matched balance, counts, selected sales, treated-control pairs, and matched pretrend tests.

data/UA_DID_analysis.csv is the bundled complete-case input. Saved results are the delivered baseline; reruns write to python_results or stata_results. For corrected or new data, rerun all relevant analyses rather than combining results from different datasets. In particular, the supplied stories values have not been newly corrected in the baseline package.

Model label reference

Label	Meaning
DID_month_FE	Unweighted conventional DID with month effects

DID_month_FE_exclude2021	Conventional sensitivity excluding 2021
DRDID_RC_locally_efficient	Dedicated doubly robust repeated cross section DID
Matching_then_DID	Matching followed by weighted conventional DID